



(Academic Session: 2024 - 2025)

ONLINE TEST SERIES CONTACT PROGRAMME

JEE (MAIN)

PHASE # 3

MINOR TEST # 8

DATE : 10-11-2024

PATTERN : MAIN

**COURSE :
ENTHUSE & LEADER**

READ THE INSTRUCTIONS CAREFULLY

Important Instructions:

1. This paper contains a total number of 75 questions.
2. The Questions appearing in the paper are based on Physics, Chemistry and Mathematics.
3. Duration of the test is 3 hours.

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Corporate Address : One Biz Square, A-12 (a), Road No. 1, Indraprastha Industrial Area, Kota - 324005 (Raj.)



+91-9513736499 |



+91-7849901001 |



wecare@allen.in |



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SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

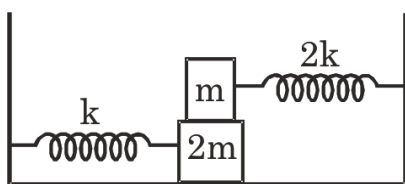
1. A particle is performing simple harmonic motion with time period T . At an instant its speed is 60% of its maximum value and is increasing. After an interval Δt its speed becomes 80% of its maximum value and is decreasing. The smallest value of Δt in terms of T will be :-

- (A) $\frac{T}{4}$ (B) $\frac{T}{2}$
(C) $\frac{3T}{8}$ (D) $\frac{3T}{4}$

2. A particle moves on the x -axis according to the equation $x = A + B \sin \omega t$. The motion is simple harmonic with amplitude :-

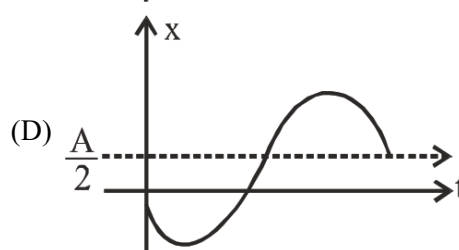
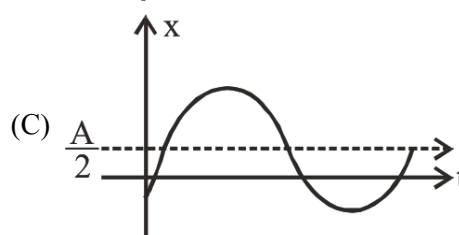
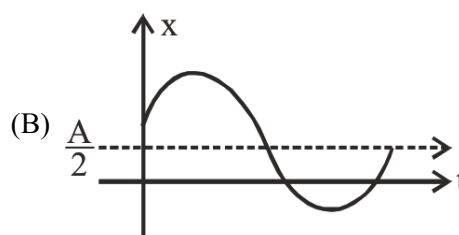
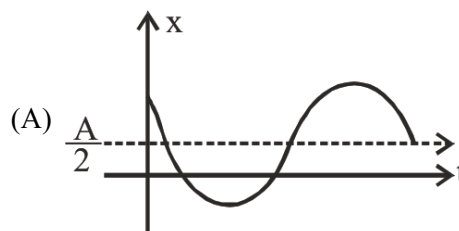
- (A) A (B) B
(C) $A + B$ (D) $\sqrt{A^2 + B^2}$

3. In the shown arrangement, the coefficient of friction between the blocks is μ and no friction between ground surface and block $2m$. The blocks are displaced slightly and released. They move together without slipping on each other. The time period will be :-



- (A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$
(C) $2\pi\sqrt{\frac{2m}{3k}}$ (D) $2\pi\sqrt{\frac{9m}{2k}}$

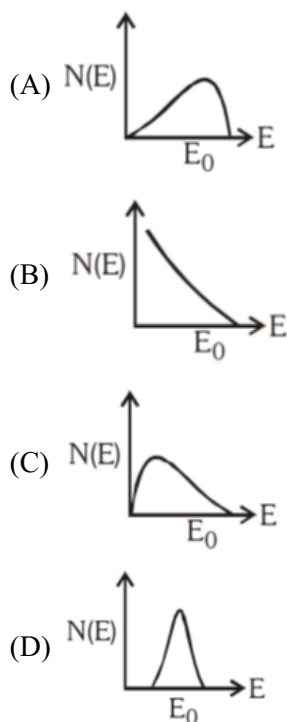
4. A particle performing S.H.M. about mean position $x = \frac{A}{2}$ and at $t = 0$. It has displacement $\frac{-3A}{4}$ and moving away from the origin. Then which of the following is its possible graph between position (x) and time (t)



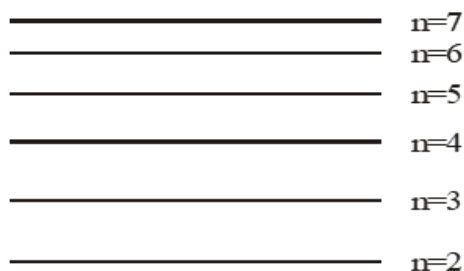
5. A particle moves along X axis such that its acceleration is given by $a = -\beta (x - 2)$. Where β is a positive constant and x is the position coordinate. The time period of oscillation (in sec) :-

- (A) π
(B) 2π
(C) $\frac{\pi}{\sqrt{\beta}}$
(D) $\frac{2\pi}{\sqrt{\beta}}$

6. Both the frequency and the intensity of a beam of light falling on the surface of photoelectric material are increased by a factor of two. This will:
- (A) increase both, the maximum kinetic energy of the photo-electrons, as well as photoelectric saturation current by a factor of two.
- (B) increase the maximum kinetic energy of the photo-electrons by a factor greater than two and would increase the photoelectric saturation current by a factor of two.
- (C) increase the maximum kinetic energy of the photoelectrons by a factor greater than two and will have no effect on the magnitude of the photoelectric saturation current produced.
- (D) increase the maximum kinetic energy of the emitted photo-electrons by a factor of two but will have no effect on the saturation photoelectric current.
7. The binding energies of deuteron (${}_1\text{H}^2$) and α -particle (${}_2\text{He}^4$) are 1.125 and 7.2 MeV/nucleon respectively. In the process ${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^4$, Amount of energy transferred is & which nucleus is more stable ?
- (A) 24.3 MeV liberated, ${}_1\text{H}^2$
- (B) 24.3 MeV liberated, ${}_2\text{He}^4$
- (C) 24.3 MeV supplied, ${}_1\text{H}^2$
- (D) 24.3 MeV supplied, ${}_2\text{He}^4$
8. A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength λ ($\lambda < \frac{hc}{\phi_0}$). The fastest photoelectron has a de Broglie wavelength λ_d . A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d . Then the ratio $\frac{\Delta\lambda_d}{\Delta\lambda}$ is proportional to
- (A) $\frac{\lambda_d^3}{\lambda^2}$ (B) $\frac{\lambda_d^3}{\lambda}$
- (C) $\frac{\lambda_d^2}{\lambda^2}$ (D) $\frac{\lambda_d}{\lambda}$
9. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{2+} is-
- (A) 30.6 eV (B) 13.6 eV
- (C) 3.4 eV (D) 122.4 eV
10. Energy levels A , B and C of a certain atom correspond to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1 , λ_2 and λ_3 are wave lengths of radiations corresponding to transitions C to B , B to A and C to A respectively, which of the following relations is correct :
- (A) $\lambda_3 = \lambda_1 + \lambda_2$ (B) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
- (C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (D) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$
11. If each fission in a U^{235} nucleus releases 200 MeV, how many fissions must occurs per second to produce a power of 1 KW
- (A) 1.325×10^{13} (B) 3.125×10^{13}
- (C) 1.235×10^{13} (D) 2.135×10^{13}
12. The energy spectrum of β -particles, number $N(E)$ as a function of β -energy E emitted from a radioactive source is-



13. The maximum number of emission lines for atomic hydrogen that you would expect to see with naked eye if the only electronic levels involved are those shown in the figure, is



- (A) 6 (B) 5 (C) 21 (D) ∞

14. A particle is travelling 4 times as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2 : 1, the mass of the particle is :-

- (A) $\frac{1}{16}$ times the mass of e^-
 (B) 8 times the mass of e^-
 (C) 16 times the mass of e^-
 (D) $\frac{1}{8}$ times the mass of e^-

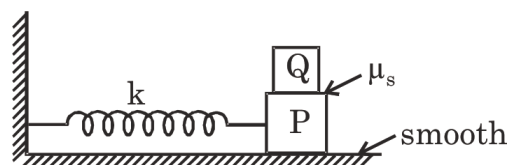
15. Which of the following transitions of He^+ ion will give rise to spectral line which has same wavelength as some spectral line in hydrogen atom ?

- (A) $n = 4$ to $n = 2$ (B) $n = 6$ to $n = 5$
 (C) $n = 6$ to $n = 3$ (D) None of these

16. A particle performs S.H.M. of amplitude A with angular frequency ω along a straight line. When it is at a distance $\frac{\sqrt{3}}{2} A$ from mean position, its kinetic energy gets increased by an amount $\frac{1}{2} m\omega^2 A^2$ due to an impulsive force. Then its new amplitude becomes-

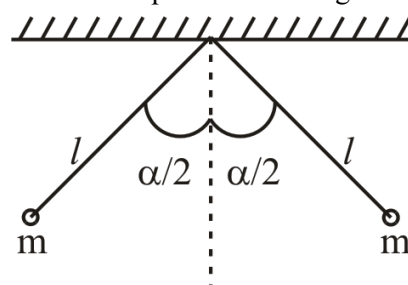
- (A) $\frac{\sqrt{5}}{2} A$ (B) $\frac{\sqrt{3}}{2} A$
 (C) $\sqrt{2} A$ (D) $\sqrt{5} A$

17. A block P of mass m is placed on a frictionless horizontal surface. Another block Q of mass $2m$ is kept on P and connected to the wall with the help of a spring of spring constant k as shown in the figure. μ_s is the coefficient of friction between P and Q. The blocks move together performing SHM of amplitude A . The maximum value of the friction force between P and Q is



- (A) $\frac{2kA}{3}$ (B) $\frac{kA}{2}$
 (C) zero (D) $\mu_s mg$

18. A light rigid wire of length 2ℓ is bent at centre, two identical particles each of mass m are attached at two ends of the wire. If angle between two halves of wire is α and wire is hinged in a vertical plane at centre, then find period of small oscillation of system in vertical plane containing the wire.



- (A) $2\pi\sqrt{\frac{\ell}{g \cos \alpha}}$ (B) $2\pi\sqrt{\frac{\ell}{g \sin \alpha}}$
 (C) $2\pi\sqrt{\frac{\ell}{g \cos(\frac{\alpha}{2})}}$ (D) $2\pi\sqrt{\frac{\ell}{g}}$

19. A free particle with initial kinetic energy E and zero potential energy and de-broglie wavelength λ , enters a region in which it has potential energy V . What is the particle's new wavelength?

- (A) $\lambda \left(1 - \frac{E}{V}\right)^{-1}$ (B) $\lambda \left(1 - \frac{V}{E}\right)^{-1}$
 (C) $\lambda \left(1 - \frac{E}{V}\right)^{-\frac{1}{2}}$ (D) $\lambda \left(1 - \frac{V}{E}\right)^{-\frac{1}{2}}$

20. In a cricket match, batsman hit the ball (mass = m) high in the air and fielding side player catch the ball near boundary. De broglie wavelength of ball just after hit from bat is λ_1 and just before the catch is λ_2 . Assume height at which ball is hit by bat and height at which ball is catch are same. Find total work done by air-resistance (Mechanical energy is loss only due to air-resistance)

- (A) $\frac{h^2}{m} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$
 (B) $\frac{h^2}{2m} \left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2} \right)$
 (C) $\frac{h^2}{2m} \left(\frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} \right)$
 (D) $\frac{h^2}{2m} \left(\frac{1}{\lambda_1} - \frac{1}{-\lambda_2} \right)$

SECTION-II : (Maximum Marks: 20)

This section contains 05 questions.

The answer to each question is a **Numerical Value**.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

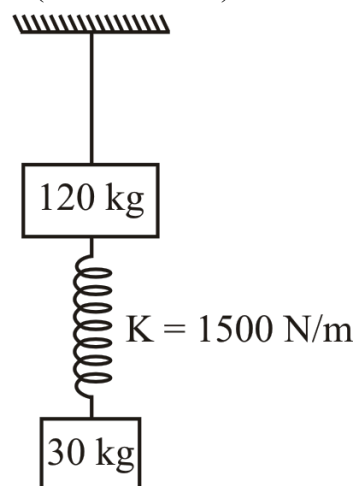
Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : -1 If wrong answer is entered.

1. Ratio of de broglie wavelength of electron in third orbit of hydrogen atom to the de broglie wavelength of electron in first orbit of hydrogen is

2. A hydrogen like atom has one electron revolving around a stationary nucleus. The energy required to excite the electron from the second orbit to the third orbit is 47.2 eV. The atomic number of the atom is ____.
3. In an experiment on photoelectric effect, the slope of the cut-off voltage (y-axis) versus frequency (x-axis) of incident light is found to be 4.12×10^{-15} V s. If the value of Planck's constant is $P \times 10^{-15}$ eV-s then value of $100 \times P \times 10^{-15}$ will be.
4. The system shown is in equilibrium. The string is cut at $t = 0$. The spring comes to its natural length at $t = t_1$ sec. If minimum value of t_1 is T then value of $5T$ will be (Take : $\pi^2 = 10$)



5. Three SHM in the same direction, having same amplitude and same period, are superimposed. If each differ in phase from the next by 45° . Then, find the energy associated with resulting motion taking energy associated with each of given SHM as 1 J. If the calculated energy is E then value of $100E$ will be.

SECTION-I : (Maximum Marks: 80)

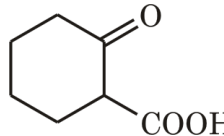
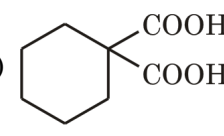
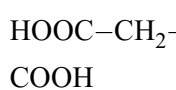
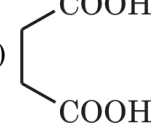
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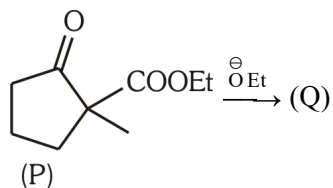
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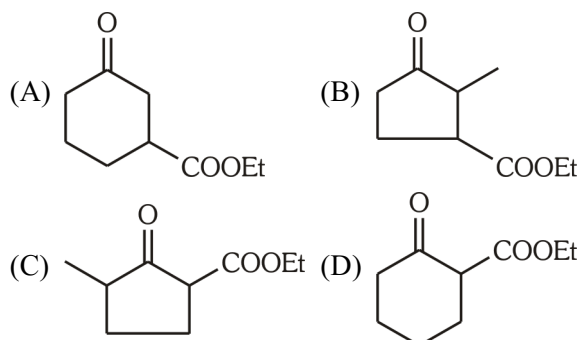
Negative Marks : -1 If wrong option is selected.

- 1 mol of nitrogen is mixed with 3 moles of hydrogen in a $\sqrt{3}$ litre container where 66.67 % of nitrogen is converted into ammonia by the following reaction : $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, then the value of K_C for the reaction $NH_3(g) \rightarrow \frac{1}{2}N_2(g) + \frac{3}{2}H_2$ will be -
(A) 2 M (B) 1/2 M
(C) 4 M (D) 1/4 M
- For the reaction $A(s) + 2B(g) \rightleftharpoons C(g) + 3D(g)$, which of the following relation is incorrect ?
(A) $K_p = K_c(RT)^1$
(B) $K_p = K_c(RT)^2$
(C) $\log \frac{K_p}{K_c} - 2 \log RT = 0$
(D) $K_c = \frac{[C]_{eq}[D]_{eq}^3}{[B]_{eq}^2}$
- $(NH_4)_2CO_3(s) \rightleftharpoons 2NH_3(g) + CO_2(g) + H_2O(g)$
The value of K_p for above equilibrium is 64 atm^4 .
The equilibrium pressure of the system is
(A) 4 atm (B) 8 atm
(C) 2 atm (D) 6 atm
- At 25°C , K_a of weak monobasic acid, HA, is 5×10^{-6} . K_b of A^- is -
(A) 5×10^{-6} (B) 2×10^5
(C) 2×10^{-9} (D) 5×10^{-8}

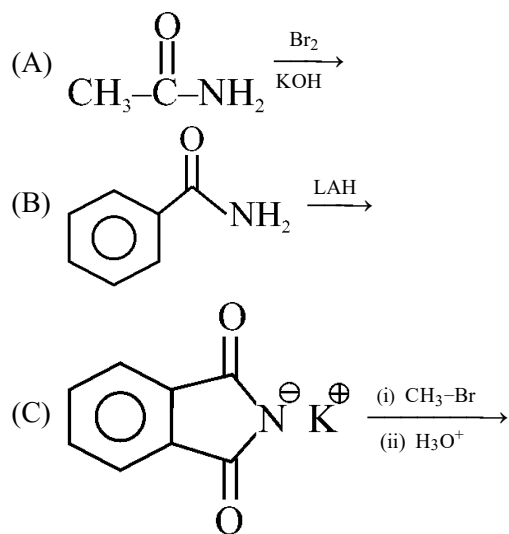
- A salt MX has $K_{sp} = 4 \times 10^{-10}$. What value of K_{sp} must another salt MX_3 have if the molar solubility of the two salts is to be identical :
(A) 3.2×10^{-10} (B) 1.024×10^{-19}
(C) 1.78×10^{-5} (D) 4.32×10^{-18}
- The compound which undergo nucleophilic substitution fastest would be :
(A) $CH_3CH_2CONH_2$
(B) $CH_3CH_2COOCH_3$
(C) CH_3CH_2COCl
(D) $CH_3CH_2COCH_3$
- Which of the following compound will not give CO_2 gas on heating ?
(A)  (B) 
(C)  (D) 
- Ester having α -H' gives β , keto ester in presence of alkoxide ion. This reaction is known as Claisen condensations reaction.



Possible structure of "Q" is

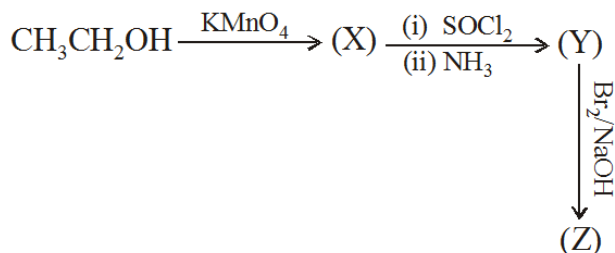


9. In which of the following reaction, one of the product obtained will give isocyanide test ?



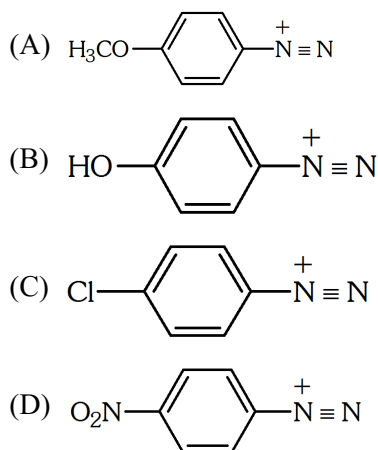
(D) All of these

10. In the following sequence of reactions 'Z' will be :

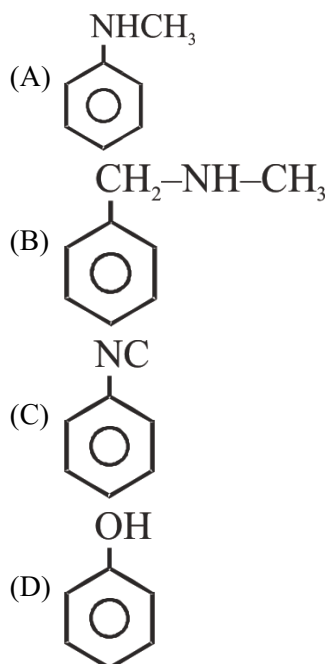


- (A) Acetic acid
(B) Acetone
(C) Methyl amine
(D) Ethyl amine

11. Which of the following is most reactive ion in azocoupling reaction ?



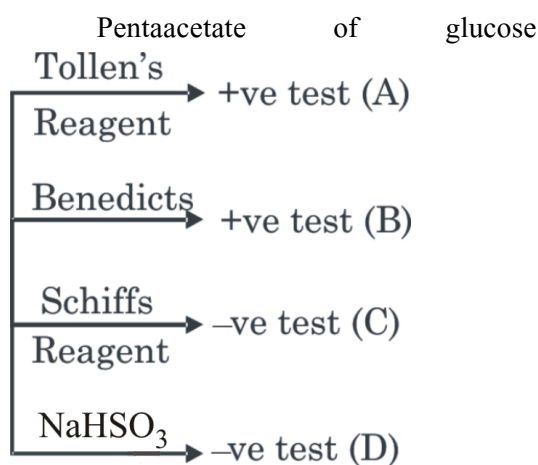
12. (Q) in following sequence is :



13. In milk sugar which linkage is presence.

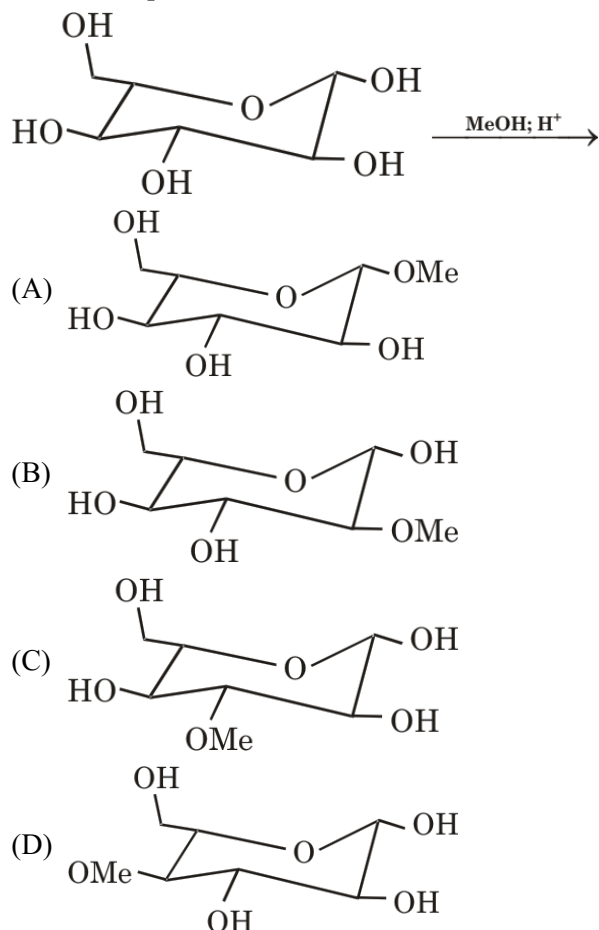
- (A) C1 – C4 α -glycosidic linkage
(B) C1 – C2 α -glycosidic linkage
(C) C1 – C4 β -glycosidic linkage
(D) C1 – C2 β -glycosidic linkage

14. Which of the following is correct.

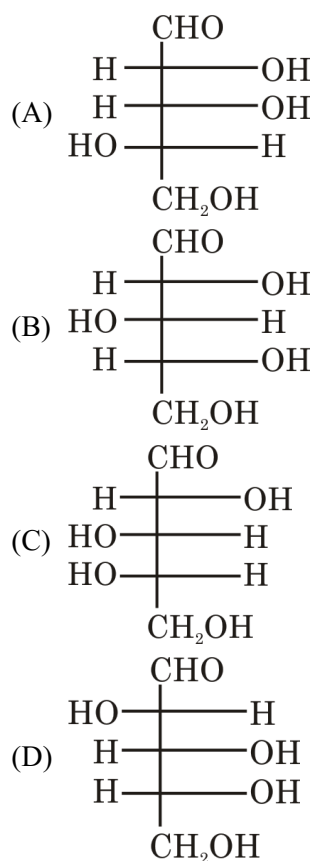


- (A) A and D are correct
(B) B and C are correct
(C) A and B are correct
(D) C and D are correct

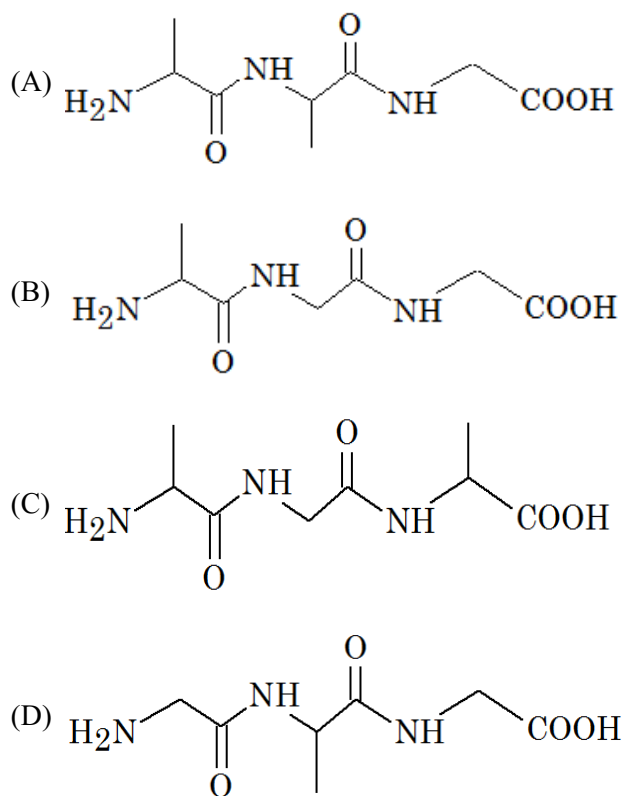
15. Predict the product



16. Which of the following gives an optically inactive aldaric acid on oxidation with dilute HNO_3 acid ?



17. The correct structure of tripeptide made up of Alanine-Glycine-Alanine is :



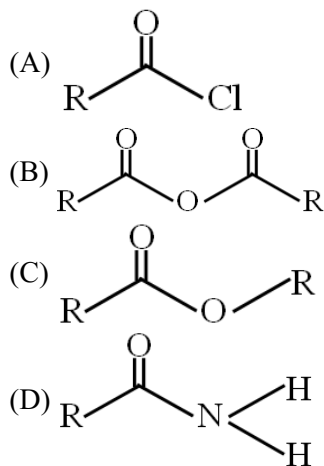
18. Acetic acid forms dimer in vapour phase due to

- (A) Intramolecular H-Bonding
(B) Inter molecular H-bonding
(C) Orbital-overlapping
(D) None of these

19. Which of the following acids has the smallest dissociation constant?

- (A) $\text{CH}_3\text{CHFCOOH}$
(B) $\text{FCH}_2\text{CH}_2\text{COOH}$
(C) $\text{BrCH}_2\text{CH}_2\text{COOH}$
(D) $\text{CH}_3\text{CHBrCOOH}$

20.



The **correct** order of their reactivity towards hydrolysis at room temperature is :

- (A) (A) > (B) > (C) > (D)
 (B) (D) > (A) > (B) > (C)
 (C) (D) > (B) > (A) > (C)
 (D) (A) > (C) > (B) > (D)

SECTION-II : (Maximum Marks: 20)

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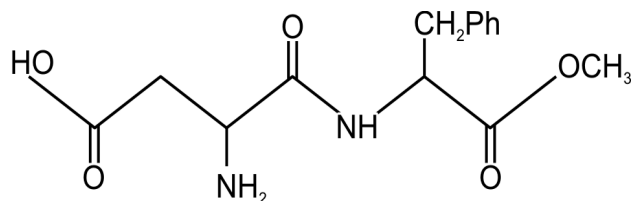
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : -1 If wrong answer is entered.

- For the reversible reaction $A(g) + B(s) \rightleftharpoons 2C(g)$, $\frac{K_p}{K_c} = (RT)^x$. Hence x is :
- Calculate the change in pressure (in atm) when 2 mole of NO and 16 g O_2 in a 6.25 litre originally at $27^\circ C$ react to produce the maximum quantity of NO_2 possible according to the equation.
 $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
 (Take $R = \frac{1}{12}$ ltr. Atm/mol K)
- The acid ionization (hydrolysis) constant of Zn^{2+} is 1.0×10^{-9} then the pH of a 0.001 M solution of $ZnCl_2$ is
- Aspartane is an artificial sugar has following structure. How many functional groups are present in the molecule.



- Identify total number of esters having molecular formula $C_6H_{12}O_2$ which give "EtOH" as one of the product on acidic hydrolysis :

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

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Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. If the lines $3x - 4y + 10 = 0$ and $3x - 4y - 15 = 0$ are tangents to a circle, then the radius of the circle is :-

(A) 5 (B) $25/2$
(C) $5/2$ (D) None of these

2. A uni-modular tangent vector on the curve $x = t^2 + 2$, $y = 4t - 5$, $z = 2t^2 - 6t$ at $t = 2$ is :-

(A) $\frac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k})$
(B) $\frac{1}{3}(\hat{i} - \hat{j} - \hat{k})$
(C) $\frac{1}{6}(2\hat{i} + \hat{j} + \hat{k})$
(D) $\frac{2}{3}(\hat{i} + \hat{j} + \hat{k})$

3. A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the co-ordinate axes. The line meets the plane $2x + y + 2 = 9$ at point Q. The length of the line segment PQ equals

(A) 3 (B) $\sqrt{2}$
(C) $\sqrt{3}$ (D) 2

4. If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then k is -

(A) 2 or $+\frac{3}{2}$ (B) $+2$ or $-\frac{3}{2}$
(C) -2 or $-\frac{3}{2}$ (D) -2 or $\frac{3}{2}$

5. Vector $\hat{a}$ in the plane of $\vec{b} = 2\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ is such that it is equally inclined to $\vec{b}$ and $\vec{d}$ where $\vec{d} = \hat{j} + 2\hat{k}$. The value of $\hat{a}$ is :-

(A) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
(B) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$
(C) $\frac{2\hat{i} + \hat{j}}{\sqrt{5}}$
(D) None

6. The locus of the mid-points of the chords of the circle $x^2 - y^2 - ax - by = 0$, which subtends a right angle at $\left(\frac{a}{2}, \frac{b}{2}\right)$ is :

(A) $ax + by = 0$
(B) $ax + by = a^2 + b^2$
(C) $x^2 - y^2 - ax - by - \frac{a^2 + b^2}{8} = 0$
(D) $x^2 + y^2 - ax - by + \frac{a^2 + b^2}{8} = 0$

7. If $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar vectors and $\vec{a} \times \vec{c}$ is perpendicular to $\vec{a} \times (\vec{b} \times \vec{c})$, then the value of $[\vec{a} \times (\vec{b} \times \vec{c})] \times \vec{c}$ is equal to :-

(A) $[\vec{a}\vec{b}\vec{c}]\vec{c}$ (B) $[\vec{a}\vec{b}\vec{c}]\vec{b}$
(C) $\vec{0}$ (D) $[\vec{a}\vec{b}\vec{c}]\vec{a}$

8. $y = mx$ is a chord of a circle of radius 'a' and the diameter of the circle lies along x-axis and one end of this chord is origin. The equation of the circle described on this chord as diameter is :-

(A) $(1 + m^2)(x^2 + y^2) - 2ax = 0$
(B) $(1 + m^2)(x^2 + y^2) - 2a(x + my) = 0$
(C) $(1 + m^2)(x^2 + y^2) + 2a(x + my) = 0$
(D) $(1 + m^2)(x^2 + y^2) - 2a(x - my) = 0$

9. Given three non-zero, non-coplanar vectors $\vec{a}, \vec{b}$ and $\vec{c}$, $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ and $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$. The vectors $\vec{r}_1 + 2\vec{r}_2$ and $2\vec{r}_1 + \vec{r}_2$ are collinear then (p, q) is :-
- (A) (0,0)
(B) (1,-1)
(C) (-1,1)
(D) (1, 1)
10. Tangents are drawn from the point (4, 3) to the circle $x^2 + y^2 = 9$. The area of the triangle formed by them and the line joining their points of contact is :-
- (A) $\frac{24}{25}$
(B) $\frac{64}{25}$
(C) $\frac{192}{25}$
(D) $\frac{192}{5}$
11. If $4\hat{i} + 7\hat{j} + 8\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $2\hat{i} + 5\hat{j} + 7\hat{k}$ are the position vectors of the vertices A, B and C, respectively, of triangle ABC, then the position vector of the point where the bisector of angle A meets BC is :-
- (A) $\frac{2}{3}(-6\hat{i} - 8\hat{j} - 6\hat{k})$
(B) $\frac{2}{3}(6\hat{i} + 8\hat{j} + 6\hat{k})$
(C) $\frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$
(D) $\frac{1}{3}(5\hat{j} + 12\hat{k})$
12. The number of integral values of λ for which $x^2 + y^2 + \lambda x + (1 - \lambda)y + 5 = 0$ is the equation of a circle whose radius cannot exceed 5, is :-
- (A) 14
(B) 18
(C) 16
(D) None of these
13. The equation of the plane passing through the lines $\frac{x-4}{1} = \frac{y-3}{1} = \frac{z-2}{2}$ and $\frac{x-3}{1} = \frac{y-2}{-4} = \frac{z}{5}$ is :-
- (A) $11x - y - 3z = 35$
(B) $11x + y - 3z = 35$
(C) $11x - y + 3z = 35$
(D) none of these
14. The cartesian equation of the plane $\vec{r} = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$ is :-
- (A) $2x + y = 5$
(B) $2x - y = 5$
(C) $2x + z = 5$
(D) $2x - z = 5$
15. Let $A \equiv (1, 2)$, $B \equiv (3, 4)$ and let $C \equiv (x, y)$ be a point such that $(x - 1)(x - 3) + (y - 2)(y - 4) = 0$. If area $(\Delta ABC) = 1$ then maximum number of positions of C in the x - y plane is :-
- (A) 2
(B) 4
(C) 8
(D) 10
16. A line segment has length 63 and dr's are 3, -2, 6. If the line makes an obtuse angle with x-axis. The components of the line vector are :-
- (A) 27, -18, 54
(B) -27, 18, -54
(C) -27, 18, 54
(D) 27, -18, -54
17. The equation of the circle passing through the point $(-1, -3)$ and touching the line $4x + 3y - 12 = 0$ at the point $(3, 0)$ is :-
- (A) $x^2 + y^2 - 2x + 3y - 3 = 0$
(B) $x^2 + y^2 + 2x - 3y - 5 = 0$
(C) $2x^2 + 2y^2 - 2x + 5y - 8 = 0$
(D) None of these

18. The equation of a line which passes through point $A(1,0,-1)$ and perpendicular to the straight lines and $\vec{r} = 2\hat{i} - \hat{j} + \hat{k} + \lambda(2\hat{i} + 7\hat{j} - 3\hat{k})$ and $\vec{r} = 3\hat{i} - \hat{j} + 3\hat{k} + \lambda(2\hat{i} - 2\hat{j} + 5\hat{k})$, is :-

- (A) $\frac{x-2}{29} = \frac{y}{-16} = \frac{z+1}{18}$
 (B) $\frac{x-1}{29} = \frac{y}{-16} = \frac{z+1}{-18}$
 (C) $\frac{x-1}{16} = \frac{y}{29} = \frac{z-1}{18}$
 (D) None of these

19. A variable circle passes through the fixed point $P(a, b)$ and touches y-axis. The locus of the other end of the diameter through P is :-

- (A) $(x-a)^2 = 4by$
 (B) $(x-b)^2 = 4ay$
 (C) $(y-a)^2 = 4bx$
 (D) $(y-b)^2 = 4ax$

20. The intercepts made on the axes by the plane which bisects the line joining the points $(1, 2, 3)$ and $(-3, 4, 5)$ at right angles are

- (A) $\left(-\frac{9}{2}, 9, 9\right)$
 (B) $\left(\frac{9}{2}, 9, 9\right)$
 (C) $\left(9, -\frac{9}{2}, 9\right)$
 (D) $\left(9, \frac{9}{2}, 9\right)$

SECTION-II : (Maximum Marks: 20)

This section contains 05 questions.

The answer to each question is a **Numerical Value**.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : -1 If wrong answer is entered.

- If the circle $x^2 + y^2 + 2ax + 2by + c = 0$ and $x^2 + y^2 + 2cx + 2ay + c = 0$ where $c > 0$, have exactly one point is common, then the value of $\frac{(a+b)^2}{2c}$ is :
- If $\vec{a} = \hat{i} - \hat{j} - \hat{k}$ and $\vec{b} = \lambda\hat{i} - 3\hat{j} + \hat{k}$ and the projection of $\vec{b}$ on $\vec{a}$ is $\frac{4}{3}(\hat{i} - \hat{j} - \hat{k})$, then λ is equal to :-
- If the line $3x - 4y - k = 0$ ($k > 0$) touches the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ at (a, b) then $k + a + b$ is equal to :-
- If the shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is $\lambda\sqrt{30}$ units then the value of λ is
- The radius of the larger circle lying in the first quadrant and touching the line $4x + 3y - 12 = 0$ and the coordinate axes is-



ONLINE TEST SERIES CONTACT PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Main)
MINOR
10-11-2024

JEE(Main) : ENTHUSE + LEADER COURSE (PHASE : 3)
ANSWER KEY

PART 1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	A	D	D	C	B	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	D	A	C	A	C	D	B
SECTION-II	Q.	1	2	3	4	5					
	A.	3	5	412	1	582					

PART 2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	C	D	C	D	C	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	D	A	B	C	B	C	A
SECTION-II	Q.	1	2	3	4	5					
	A.	1	2	6	4	2					

PART 3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	B	B	D	C	B	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	C	B	B	A	B	D	A
SECTION-II	Q.	1	2	3	4	5					
	A.	1	2	20	3	6					

HINT – SHEET

PART 1 : PHYSICS

SECTION-I

1. **Ans (A)**

Ist case $\rightarrow n = 60^\circ$ of $v_{\max}$

$$v = \frac{60 \times A\omega}{100}$$

$$n = 0.6 A\omega$$

$$\omega\sqrt{A^2 + x_1^2} = 0.6A\omega$$

$$A^2 + x_1^2 = 0.36A^2$$

$$x_1^2 = 0.64A^2$$

$$x_1 = 0.8A$$

IInd case $\rightarrow n = 80\%$ of $v_{\max}$

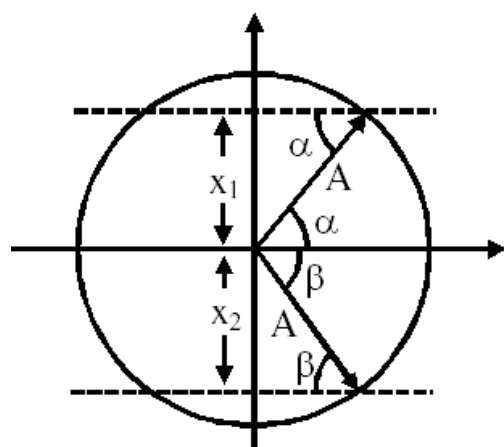
$$v = \frac{80 \times A\omega}{100} = 0.8A\omega$$

$$\omega\sqrt{A^2 - x_2^2} = 0.8A\omega$$

$$\omega^2(A^2 - x_2^2) = 0.64A^2\omega^2$$

$$x_2^2 = 0.36A^2$$

$$x_2 = 0.6A$$



$$\sin \alpha = \frac{x_1}{A} = \frac{0.8A}{A}$$

$$\sin \alpha = 0.8 = \frac{4}{5} = \sin 53^\circ$$

$$\alpha = 53^\circ$$

$$\sin \beta = \frac{x_2}{A} = \frac{0.6A}{A}$$

$$\sin \beta = 0.6 = \frac{3}{5} = \sin 37^\circ$$

$$\beta = 37^\circ$$

$$\theta = \omega t$$

$$(\alpha + \beta) = \omega t \Rightarrow t = \frac{90^\circ}{\omega} = \frac{\pi \times T}{2 \times 2\pi} = \frac{T}{4}$$

2. **Ans (B)**

$$(x - A) = B\sin \omega t$$

SHM of particle about $x = A$ and its amplitude is B .

3. **Ans (A)**

$$T = 2\pi\sqrt{\frac{M_{\text{total}}}{k_{\text{eff}}}}$$

$$k_{\text{eff}} = k_1 + k_2 \quad \text{Parallel combination}$$

$$= 2k + k = 3k$$

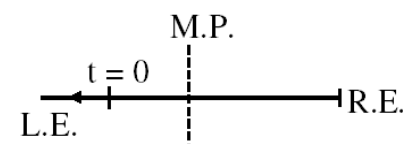
$$T = 2\pi\sqrt{\frac{3m}{3k}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

4. **Ans (D)**

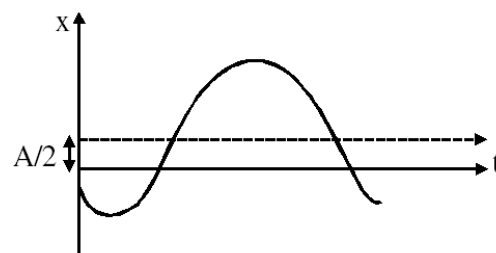
$$\text{Mean position at } x = \frac{A}{2}$$

$$\text{at } t = 0 \quad x = -\frac{3A}{4}$$



at $t = 0$ it will move towards left extreme

(negative displacement)



5. **Ans (D)**

$a = -\beta (x - 2)$ Compares with the standard equation of SHM $a = -\omega^2 x$.

$$\therefore \omega = \sqrt{\beta} \quad (\text{Equilibrium position } x = 2)$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\beta}}$$

6. **Ans (C)**

The maximum K.E. of ejected photoelectron

$$\text{is } (K.E.)_{\max} = h\nu - \phi_0$$

If the frequency of photon is doubled,

maximum kinetic energy of photon electron

$$\text{becomes } (K.E.)'_{\max} = 2h\nu - \phi_0$$

$$\frac{(K.E.)'_{\max}}{(K.E.)_{\max}} = \frac{2(h\nu - \frac{\phi_0}{2})}{h\nu - \phi_0} > 2$$

$$\text{Photo current} \propto \frac{\text{intensity of beam}}{h\nu}$$

If intensity and frequency both are doubled, the photocurrent remains same.

7. **Ans (B)**

$$\Delta Q = BE_p - BE_R$$

$$= 4 \times 7.2 - 4 \times 1.125$$

$$= 28.8 - 4.5 = 24.3 \text{ MeV}$$

8. **Ans (A)**

$$\frac{hc}{\lambda} - \phi = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$p = \frac{h}{\lambda_d}$$

$$\frac{hc}{\lambda} - \phi = \frac{h^2}{2m\lambda_d^2}$$

$$-hc \frac{\Delta\lambda}{\lambda^2} = \frac{h^2}{2m} \frac{(-2)}{\lambda_d^3} \Delta\lambda_d$$

$$-\frac{hc \cdot 2m}{h^2(-2)} = \frac{\lambda_d^3}{\lambda^2} = \frac{\Delta\lambda_d}{\Delta\lambda}$$

9. **Ans (A)**

$$E = 13.6 \times \frac{9}{4} = 30.6 \text{ eV}$$

10. **Ans (B)**

$$E_C - E_B = \frac{hc}{\lambda_1} \quad \dots(1)$$

$$E_B - E_A = \frac{hc}{\lambda_2} \quad \dots(2)$$

$$E_C - E_A = \frac{hc}{\lambda_3} \quad \dots(3)$$

On add equation (1) and (2)

$$E_C - E_A = hc \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right)$$

$$\frac{hc}{\lambda_3} = hc \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right) \Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

11. **Ans (B)**

Let n be number of fissions per second.

Each fission produces 200 MeV .

$n \times 200 \times 10^6 \text{ eV}$ is produced in one second by n fissions

$$\text{But } 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Hence, power produce

$$n \times 200 \times 10^6 \times 1.6 \times 10^{-19} \text{ Joule per second.}$$

Also $1 \text{ J/s} = 1 \text{ W}$ and

$$1000 \text{ J/s} = 1 \text{ kW.}$$

The required power is 1 kW .

$$\text{Hence, } \frac{n \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{10^3} = 1$$

$$n = \frac{10^3}{2 \times 1.6 \times 10^{-11}} = \frac{10^{14}}{3.2}$$

$$= \frac{10}{3.2} \times 10^{13} = 3.125 \times 10^{13}$$

12. **Ans (C)**

The range of β particle lie between zero to some maximum value. During beta decay,

electrons or photons are released. Because a neutron or an antineutron is emitted simultaneously, there is an emission spectrum of electrons or positrons depending on the ratio or Q reaction energies boron by the large particle. The shape of this energy curve can be predicted from the Fermi theory of beta decay.

13. **Ans (B)**

Spectral lines of Balmer series are in visible range.

Balmer series lines are obtained due to transition of an electron to $n = 2$ state from any higher state.

14. **Ans (D)**

$$\lambda = \frac{h}{p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{p_e}{p_p} = \frac{m_e v_e}{m_p v_p}$$

$$2 = \frac{m_e}{m_p} \left(\frac{v_e}{4v_e} \right)$$

$$\therefore m_p = \frac{m_e}{8}$$

15. Ans (A)

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad z = 2$$

$$\frac{1}{\lambda} = 4R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{\frac{n_1^2}{4}} - \frac{1}{\frac{n_2^2}{4}} \right)$$

$$\frac{n_1^2}{4} \text{ \& } \frac{n_2^2}{4} \text{ must be integer \& square terms}$$

$$n = 4 \text{ to } n = 2$$

16. Ans (C)

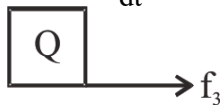
Total energy of SHM (After impulse)

$$m\omega^2 A^2 = \frac{1}{2} m\omega^2 A_1^2$$

$$\therefore A_1 = \sqrt{2}A$$

17. Ans (A)

$$-kx = \frac{3md^2x}{dt^2} \quad \dots (i)$$



$$f_3 = \frac{2md^2x}{dt^2} \quad \dots (ii)$$

18. Ans (C)

$$T = 2\pi \sqrt{\frac{l}{Mgd}} = 2\pi \sqrt{\frac{2ml\alpha}{2mg \cos \frac{\alpha}{2}}}$$

$$= 2\pi \sqrt{\frac{l}{g \cos \left(\frac{\alpha}{2} \right)}}$$

19. Ans (D)

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda' = \frac{h}{\sqrt{2m(E-V)}}$$

20. Ans (B)

$$\text{Kinetic energy of ball} = \frac{h^2}{2m\lambda^2}$$

$$\text{Loss in kinetic energy} = \frac{h^2}{2m} \left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2} \right)$$

PART 1 : PHYSICS

SECTION-II

1. Ans (3)

$$\lambda = \frac{h}{mv}$$

$$\text{velocity of an } e^- \text{ in } \propto \frac{Z}{h}$$

H-like atom

$$\lambda \propto \frac{1}{v} \text{ so } \lambda \propto \frac{k}{v} \quad (k = \text{constant})$$

$$\frac{\lambda_1}{\lambda_3} = \frac{V_3}{V_1}$$

$$\frac{\lambda_1}{\lambda_3} = \frac{Z_H}{n_3} \times \frac{n_1}{Z_H} = \frac{1}{3} \Rightarrow \frac{\lambda_3}{\lambda_1} = \frac{3}{1}$$

2. Ans (5)

$$13.6 Z^2 \left[\frac{1}{4} - \frac{1}{9} \right] = 47.2$$

$$Z^2 \approx 25$$

$$Z \approx 5.00$$

3. Ans (412)

$$eV_s = hf - \phi$$

$$v_s = \frac{h}{e} f - \phi$$

$$\text{Slope} = \frac{h}{e} = 4.12 \times 10^{-15}$$

$$h = 4.12 \times 10^{-15} \text{ eVs}$$

4. Ans (1)

$$T = 2\pi \sqrt{\frac{\mu}{K}}$$

$$\& t_1 = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \dots$$

5. Ans (582)

$$y_{\text{net}} = y_1 + y_2 + y_3$$

$$= a [\sin \omega t + \sin(\omega t + 45^\circ) + \sin(\omega t + 90^\circ)]$$

$$= (\sqrt{2} + 1) a \sin(\omega t + 45^\circ)$$

Energy ratio is

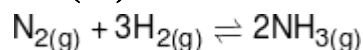
$$\frac{E_{\text{resultant}}}{E_{\text{ring}}} = \left(\frac{A}{a} \right)^2 = (\sqrt{2} + 1)^2 = (3 + 2\sqrt{2})$$

$$E_{\text{resultant}} = 5.82 \text{ J}$$

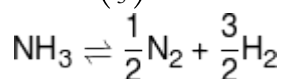
PART 2 : CHEMISTRY

SECTION-I

1. **Ans (D)**



$$K_C = \frac{\left(\frac{4}{3}\right)^2}{\left(\frac{1}{3}\right)(1)} \times (\sqrt{3})^2 = 16$$



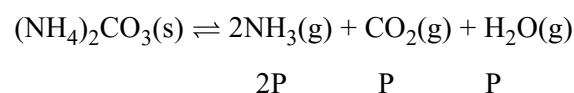
$$K_C^1 = \frac{1}{4}$$

2. **Ans (A)**

$$K_P = K_C (RT)^{\Delta n_g}$$

here $\Delta n_g = 2$

3. **Ans (B)**



$$K_P = (2P)^2 \cdot P \cdot P = 64 \text{ atm}$$

$$P = 2 \text{ atm}$$

$$P_{\text{total}} = 2P + P + P = 8 \text{ atm}$$

4. **Ans (C)**

$$K_a \times K_b = K_w$$

5. **Ans (D)**

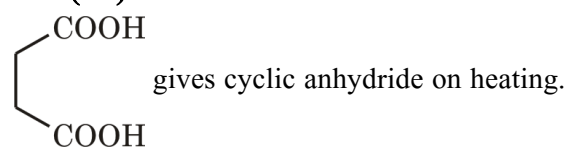
$$\text{Solubility of MX} = (4 \times 10^{-10})^{1/2}$$

$$\begin{aligned} \text{MX}_3 : K_{sp} &= 27s^4 \\ &= 27 (2 \times 10^{-5})^4 \\ &= 4.32 \times 10^{-18} \end{aligned}$$

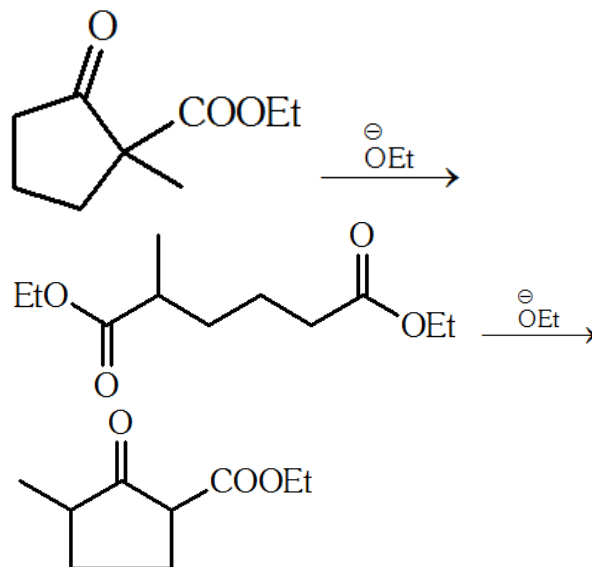
6. **Ans (C)**

Acid halide is most reactive due to -I of Cl.

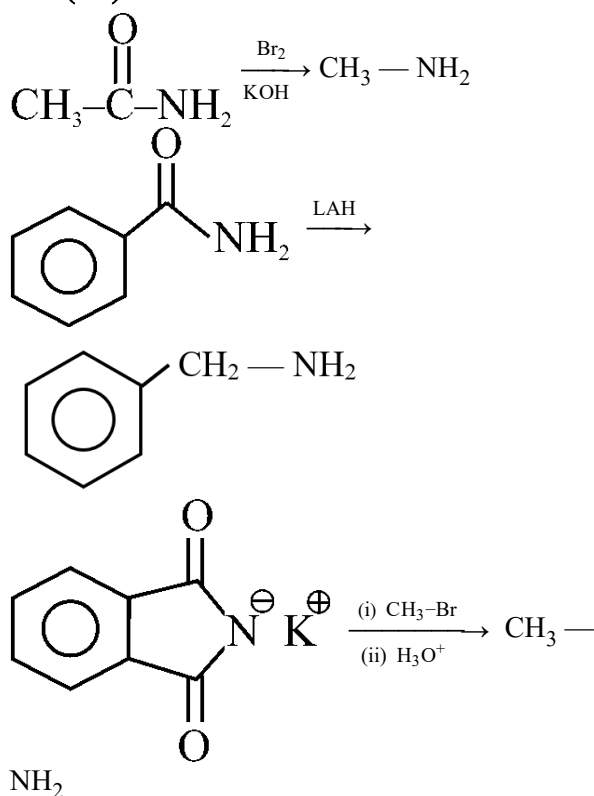
7. **Ans (D)**



8. **Ans (C)**



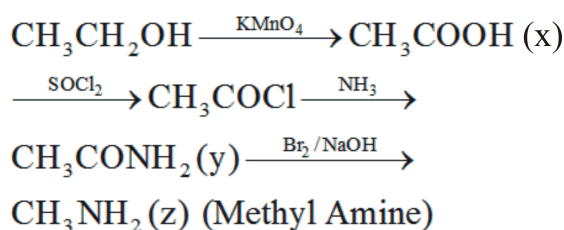
9. **Ans (D)**



NH₂

All primary amines give isosynide test

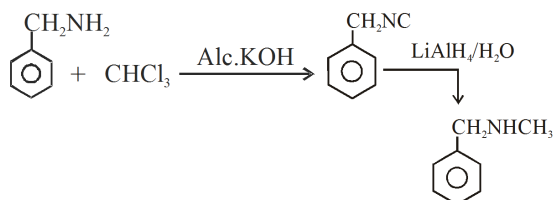
10. **Ans (C)**



11. **Ans (D)**

In azo-coupling reaction diazonium ion acts as an electrophile. The presence of NO_2 group makes the diazo group more electron deficient.

12. **Ans (B)**



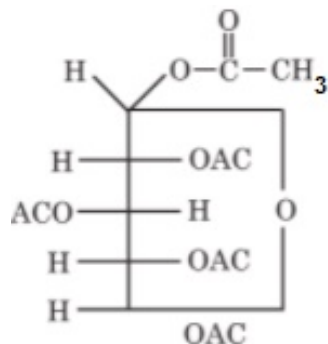
13. **Ans (C)**

Milk sugar (lactose) C1 - C4 β -glycosidic linkage is present.

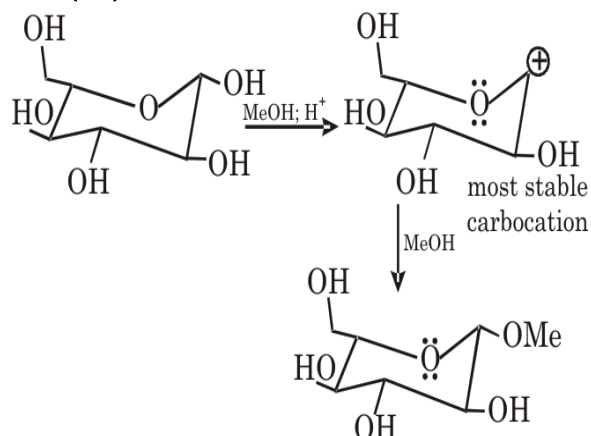
14. **Ans (D)**

There is no hemiacetal linkage in pentacetate of glucose. So, No -CHO group in alkaline medium.

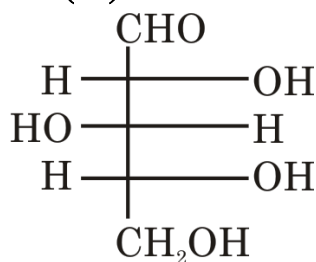
Pentacetate of glucose is



15. **Ans (A)**

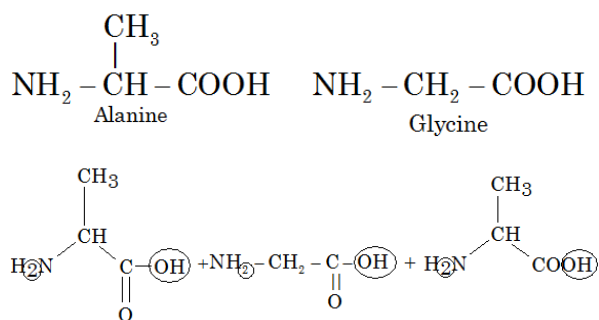


16. **Ans (B)**

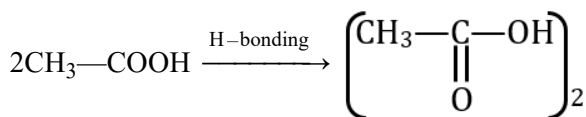


is a meso compound.

17. **Ans (C)**



18. **Ans (B)**

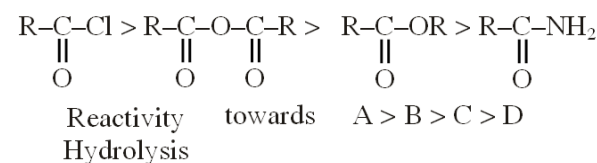


19. **Ans (C)**

k_a $\propto$ acid strength

$$\text{Acid strength} \propto \frac{-M, -H, -I}{+M, +H, +I}$$

20. **Ans (A)**



PART 2 : CHEMISTRY

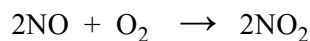
SECTION-II

1. **Ans (1)**

$$\Delta n_g = 1$$

$$\text{So, } \frac{K_P}{K_C} = (RT)^1$$

2. Ans (2)



$$\begin{array}{ccc} 2 & 0.5 & 0 \\ 2-1 & 0 & 1 \end{array}$$

$$n_i = 2.5, n_f = 2$$

$$\Delta n = 0.5$$

$$\Delta P = \frac{\Delta nRT}{v} = \frac{0.5 \times 0.0821 \times 300}{6.25} = 2$$

3. Ans (6)

$$K_h = \frac{x^2}{C} \Rightarrow x = [\text{H}^+] = \sqrt{K_h \cdot C}$$

$$[\text{H}^+] = \sqrt{10^{-9} \times 0.001} = 10^{-6} \text{ M}$$

$$\text{pH} = 6$$

4. Ans (4)

Functional group present are

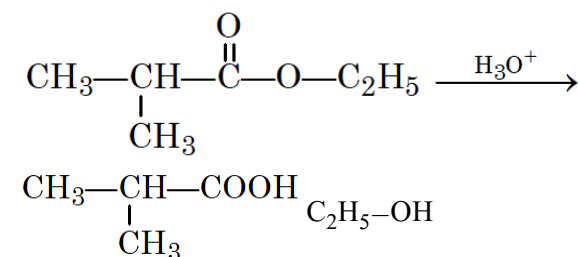
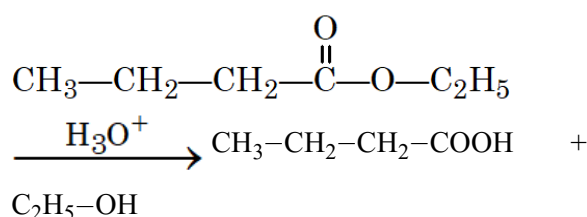
—COCH₃ (ester)

—COOH (Acid)

—NH₂ (Primary Amine)

—CONH₂ (Amide)

5. Ans (2)



PART 3 : MATHEMATICS

SECTION-I

1. Ans (C)

Given lines are parallel

Hence dist. between lines = diameter of circle

$$\text{Hence } r = \frac{1}{2} (\text{dist. between lines})$$

$$= \frac{1}{2} \left| \frac{10 - (-15)}{\sqrt{3^2 + 4^2}} \right| = \frac{5}{2}$$

2. Ans (A)

The position vector of any point at t is

$$\vec{r} = (2 + t^2)\hat{i} + (4t - 5)\hat{j} + (2t^2 - 6t)\hat{k}$$

$$\Rightarrow \frac{d\vec{r}}{dt} = 2t\hat{i} + 4\hat{j} + (4t - 6)\hat{k}$$

$$\Rightarrow \left. \frac{d\vec{r}}{dt} \right|_{t=2} = 4\hat{i} + 4\hat{j} + 2\hat{k}$$

$$\text{and } \left| \left. \frac{d\vec{r}}{dt} \right|_{t=2} \right| = \sqrt{16 + 16 + 4} = 6$$

Hence, the required unit tangent vector at t = 2 is

$$\frac{1}{3} (2\hat{i} + 2\hat{j} + \hat{k})$$

3. Ans (C)

Let $\alpha, \beta, \gamma \rightarrow$ angles made by direction continues with x, y, z - axis $\alpha = \beta = \gamma$

$$\alpha = \beta = \gamma$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$3\cos^2 \alpha = 1; \cos \alpha = \pm \frac{1}{\sqrt{3}}$$

$$p = (2, -1, 2)$$

$$\text{Equation of line } \frac{x-2}{\frac{1}{\sqrt{3}}} = \frac{y+1}{\frac{1}{\sqrt{3}}} = \frac{z-2}{\frac{1}{\sqrt{3}}} = k$$

$$Q^x = \left(\frac{k}{\sqrt{3}} + 2, \frac{k}{\sqrt{3}} - 1, \frac{k}{\sqrt{3}} + 2 \right)$$

$$2 \left(\frac{k}{\sqrt{3}} + 2 \right) + \frac{k}{\sqrt{3}} - 1 + \frac{k}{\sqrt{3}} + 2 = 9$$

$$\frac{4k}{\sqrt{3}} + 5 = 9$$

$$\frac{4k}{\sqrt{3}} = 4 \Rightarrow k = \sqrt{3}$$

$$\text{Point } Q \equiv (3, 0, 3)$$

$$P = (2, -1, 2)$$

$$PQ = \sqrt{1 + 1 + 1} = \sqrt{3}$$

4. Ans (B)

Given circles will intersect orthogonally,

$$\text{if } 2(1 \times 0 + k \times k) = 6 + k$$

$$[\text{Using : } 2(g_1g_2 + f_1f_2) = c_1 + c_2]$$

$$\Rightarrow 2k^2 - k - 6 = 0$$

$$\Rightarrow (2k + 3)(k - 2) = 0$$

$$\Rightarrow k = 2, -\frac{3}{2}$$

5. Ans (B)

Let $\vec{a} = \lambda\vec{b} + \mu\vec{c}$

$\vec{a}$ is equally inclined to $\vec{b}$ and $\vec{d}$, where

$\vec{d} = \hat{j} + 2\hat{k}$.

$$\Rightarrow \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a} \cdot \vec{d}}{|\vec{a}||\vec{d}|}$$

$$\Rightarrow \frac{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{b}}{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{d}} = \frac{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{b}}{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{d}}$$

$$= \frac{\left[\lambda(2\hat{i} + \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k}) \right] \cdot (2\hat{i} + \hat{j})}{\left[\lambda(2\hat{i} + \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k}) \right] \cdot (\hat{j} + 2\hat{k})}$$

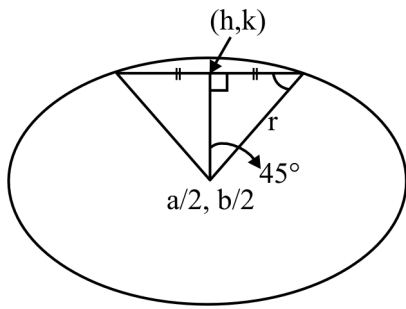
$$= \frac{\sqrt{5}}{\sqrt{5}}$$

or $\lambda(4 + 1) + \mu(2 - 1) = \lambda(1) + \mu(-1 + 2)$

or $4\lambda = 0$, i.e., $\lambda = 0$

$\therefore \hat{a} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

6. Ans (D)



$e = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2}$

$\cos 45^\circ = \frac{\sqrt{\left(4 - \frac{a}{2}\right)^2 + \left(k - \frac{b}{2}\right)^2}}{\frac{\sqrt{a^2 + b^2}}{2}}$

$\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{4} \left(\frac{(2x - a)^2 + (2y - b)^2}{\frac{(a^2 + b^2)}{4}} \right)$

$\frac{a^2 + b^2}{2} = 4x^2 + 4y^2 - 4ax - 4by + a^2 + b^2$

$4x^2 + 4y^2 - 4ax - 4by + \frac{a^2 + b^2}{2} = 0$

$x^2 + y^2 - ax - by + \frac{a^2 + b^2}{8} = 0$

7. Ans (C)

Given that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar. Thus,

$[\vec{a} \vec{b} \vec{c}] \neq 0 \quad \dots(i)$

Again $\vec{a} \times (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{c}) = 0$

or $[(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}] \cdot (\vec{a} \times \vec{c}) = 0$

or $(\vec{a} \cdot \vec{c})[\vec{b} \vec{a} \vec{c}] = 0$

or $(\vec{a} \cdot \vec{c}) = 0$

Hence, $\vec{a}$ and $\vec{c}$ are perpendicular (ii)

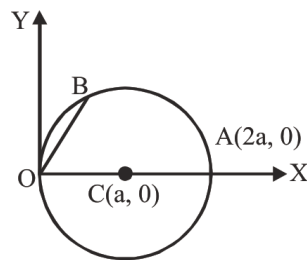
$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

or $[\vec{a} \times (\vec{b} \times \vec{c})] \times \vec{c} = \vec{0}$

8. Ans (B)

Here the equation of circle is

$(x - a)^2 + (y - 0)^2 = a^2 \Rightarrow x^2 + y^2 - 2ax = 0$



Now the point of intersection of circle and chord i.e.,

Put $y = mx$ in equation of circle and solve it.

O and B are $O(0, 0)$ and $B\left(\frac{2a}{1 + m^2}, \frac{2am}{1 + m^2}\right)$.

Hence the equation of circle (as chord OB as diameter) is $(x^2 + y^2)(1 + m^2) - 2a(x + my) = 0$.

9. Ans (D)

Let $\vec{a} = \vec{i}$, $\vec{b} = \vec{j}$, $\vec{c} = \vec{k}$

$$\begin{cases} \vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c} = p\vec{i} + q\vec{j} + \vec{k} \\ \vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c} = \vec{i} + p\vec{j} + q\vec{k} \end{cases}$$

$$\begin{cases} \vec{r}_1 + 2\vec{r}_2 = \hat{i}(p + 2) + \hat{j}(q + 2p) + \hat{k}(1 + 2q) \\ 2\vec{r}_1 + \vec{r}_2 = (2p + 1)\hat{i} + (2q + p)\hat{j} + (2 + q)\hat{k} \end{cases}$$

$$(\vec{r}_1 + 2\vec{r}_2) \parallel (2\vec{r}_1 + \vec{r}_2)$$

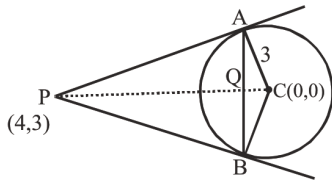
$$\Rightarrow \frac{p + 2}{2p + 1} = \frac{q + 2p}{2q + p} = \frac{1 + 2q}{2 + q}$$

$p = q$ & $q = 1$

10. Ans (C)

Equation of AB is $4x + 3y = 9$ (i)

{ $\because$ it is chord of contact }



$OQ = \frac{9}{5}$ [perpendicular distance of AB from origin]

$$AQ = \sqrt{OA^2 - OQ^2} = \sqrt{9 - \frac{81}{25}} = \frac{12}{5}$$

$$AB = 2AQ = \frac{24}{5}$$

$$PQ = \frac{16 + 9 - 9}{\sqrt{16 + 9}} = \frac{16}{5}$$

$$\text{Hence the area} = \frac{1}{2} \times \frac{24}{5} \times \frac{16}{5} = \frac{192}{25}$$

$$\text{Aliter : Required area} = \frac{a}{h^2 + k^2} (h^2 + k^2 - a^2)^{3/2}$$

$$= \frac{3}{4^2 + 3^2} (42 + 32 - 9)^{3/2} = \frac{192}{25}$$

11. Ans (C)

Suppose the bisector of angle A meets BC at D.

Then AD divides BC in the ratio AB : AC.

So, P.V. of D is given by

$$\frac{|\vec{AB}|(2\hat{i} + 5\hat{j} + 7\hat{k}) + |\vec{AC}|(2\hat{i} + 3\hat{j} + 4\hat{k})}{|\vec{AB}| + |\vec{AC}|}$$

$$\text{But } \vec{AB} = -2\hat{i} - 4\hat{j} - 4\hat{k}$$

$$\text{and } \vec{AC} = -2\hat{i} - 2\hat{j} - \hat{k}$$

$$\Rightarrow |\vec{AB}| = 6 \text{ and } |\vec{AC}| = 3$$

Therefore, P.V. of D is given by

$$\frac{6(2\hat{i} + 5\hat{j} + 7\hat{k}) + 3(2\hat{i} + 3\hat{j} + 4\hat{k})}{6 + 3}$$

$$= \frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$$

12. Ans (C)

radius ≤ 5

$$\sqrt{\left(\frac{\lambda}{2}\right)^2 + \left(\frac{1-\lambda}{2}\right)^2} - 5 \leq 5$$

$$\Rightarrow 2\lambda^2 - 2\lambda - 119 \leq 0$$

$$\Rightarrow \frac{1 - \sqrt{239}}{2} \leq \lambda \leq \frac{1 + \sqrt{239}}{2}$$

$$\Rightarrow -7.2 \leq \lambda \leq 8.2 \text{ (nearly)}$$

$$\Rightarrow \lambda = -7, -6, \dots, 8$$

13. Ans (D)

Here, the required plane is

$$a(x - 4) + b(y - 3) + c(z - 2) = 0$$

$$\text{Also } a + b + 2c = 0 \text{ and } a - 4b + 5c = 0$$

Solving, we have

$$\frac{a}{5 + 8} = \frac{b}{2 - 5} = \frac{c}{-4 - 1} = k$$

$$\frac{a}{13} = \frac{b}{-3} = \frac{c}{-5} = k$$

Therefore, the required equation of plane is

$$-13x + 3y + 5z + 33 = 0$$

14. Ans (C)

$$\vec{r} = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$$

$\Rightarrow$

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} - 2\hat{k}) + \mu(-\hat{i} + 2\hat{k})$$

which is a plane passing through

$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and parallel to the vectors

$\vec{b} = \hat{i} - \hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{k}$ therefore, it

perpendicular to the vector

$$\vec{n} = \vec{b} \times \vec{c} = -2\hat{i} - \hat{k}$$

Hence, its vector equation is

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

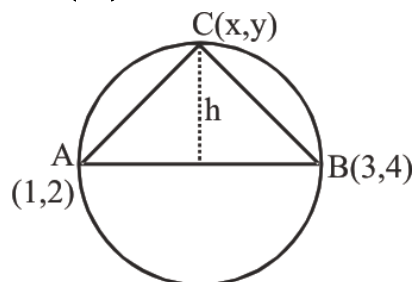
$$\Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \Rightarrow \vec{r} \cdot (-2\hat{i} - \hat{k}) = -2 - 3$$

$$\Rightarrow \vec{r} \cdot (2\hat{i} + \hat{k}) = 5$$

so, the cartesian equation is

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (2\hat{i} + \hat{k}) = 5 \text{ or } 2x + z = 5$$

15. Ans (B)



$\because$ A & B are end's of diameter, diameter $= 2\sqrt{2}$

$\therefore$ radius $= \sqrt{2}$

Let height of ΔABC is h.

$$\text{Now, } \frac{1}{2} \cdot \text{Base} \times h = 1 \quad \{\text{Base} = \text{diameter}\}$$

$$\Rightarrow \frac{1}{2} \times 2\sqrt{2} \times h = 1$$

$$\Rightarrow h = \frac{1}{\sqrt{2}}$$

$\because h < r$

therefore no. of position of C is 4.

16. Ans (B)

Let the components of the line segment vector be a, b, c then

$$a^2 + b^2 + c^2 = (63)^2 \quad \dots(i)$$

$$\text{also. } \frac{a}{3} = \frac{b}{-2} = \frac{c}{6} = \lambda \text{ (say)}$$

$$\Rightarrow a = 3\lambda, b = -2\lambda, c = 6\lambda$$

$$\text{from (i)} \quad 9\lambda^2 + 4\lambda^2 + 36\lambda^2 = (63)^2$$

$$\Rightarrow 49\lambda^2 = (63)^2$$

$$\Rightarrow \lambda = \pm \frac{63}{7} = \pm 9$$

$$\because a = 3\lambda < 0$$

$\therefore$ line makes obtuse angle with x-axis

$$\Rightarrow \lambda = -9$$

So. reqd. components are $-27, 18, -54$

17. Ans (A)

Let the equation be $x^2 + y^2 + 2gx + 2fy + c = 0$ it passes through $(-1, -3)$ and $(3, 0)$ therefore

$$10 - 2g - 6f + c = 0 \quad \dots(i)$$

$$9 + 6g + c = 0 \quad \dots(ii)$$

Slope of tangent $= -4/3$

$$\left(\frac{0+f}{3+g} \right) \left(\frac{-4}{3} \right) = -1$$

$$\Rightarrow 3g - 4f + 9 = 0 \quad \dots(iii)$$

$$\text{solving } g = -1, f = 3/2, c = -3$$

18. Ans (B)

Let the equation of line passes through

$$\frac{x-1}{a} = \frac{y-0}{b} = \frac{z+1}{c} \quad \dots(1)$$

It is perpendicular to two given lines

$$\therefore 2a + 7b - 3c = 0 \quad \dots(2)$$

$$\text{and } 2a - 2b + 5c = 0 \quad \dots(3)$$

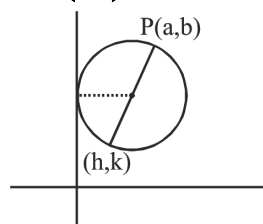
from (2) & (3)

$$\frac{a}{29} = \frac{b}{-16} = \frac{c}{-18}$$

$\therefore$ Equation of line

$$\frac{x-1}{29} = \frac{y-0}{-16} = \frac{z+1}{-18}$$

19. Ans (D)



$$\text{centre } \left(\frac{h+a}{2}, \frac{k+b}{2} \right)$$

$$\text{radius} = \frac{1}{2} \sqrt{(h-a)^2 + (k-b)^2}$$

Acc. to condition

$$\frac{1}{2} \sqrt{(h-a)^2 + (k-b)^2} = \frac{h+a}{2}$$

$$\Rightarrow (k-b)^2 = 4ah$$

$$\Rightarrow (y-b)^2 = 4ax$$

20. Ans (A)

$$\text{Let plane is } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots(i)$$

mid point of $P(1, 2, 3)$ and $Q(-3, 4, 5)$

i.e., $-1, 3, 4$ lie on Eq. (i)

$$\therefore -\frac{1}{a} + \frac{3}{b} + \frac{4}{c} = 1 \quad \dots(ii)$$

Also, PQ is parallel to normal of the plane (i)

$$\frac{1/a}{-4} = \frac{1/b}{2} = \frac{1/c}{2}$$

$$\Rightarrow \frac{1}{-2a} = \frac{1}{b} = \frac{1}{c} = \lambda \text{ (say)}$$

$$\therefore \frac{1}{a} = -2\lambda, \frac{1}{b} = \lambda, \frac{1}{c} = \lambda$$

$$\therefore \text{From Eq. (ii), } 2\lambda + 3\lambda + 4\lambda = 1$$

$$\therefore \lambda = \frac{1}{9}$$

$$a = -\frac{1}{2\lambda}, b = \frac{1}{\lambda}, c = \frac{1}{\lambda}$$

$$a = -\frac{9}{2}, b = 9, c = 9$$

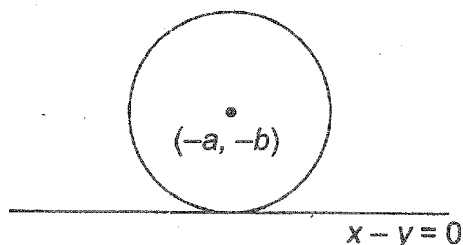
$$\text{Intercepts are } \left(-\frac{9}{2}, 9, 9 \right).$$

PART 3 : MATHEMATICS

SECTION-II

1. Ans (1)

Radical axis, $x - y = 0$



Centre of 1st circle is $(-a, -b)$ and $r = \sqrt{a^2 + b^2 - c}$.

Now perpendicular distance from center on radical axis = radius of the circle

$$(a - b)^2 = 2[a^2 + b^2 - c]$$

$$\Rightarrow a^2 + b^2 - 2ab = 2[a^2 + b^2 - c]$$

$$a^2 + b^2 + 2ab = 2c$$

$$\Rightarrow \frac{(a + b)^2}{2c} = 1$$

2. Ans (2)

$$\text{Given, } \frac{(\vec{b} \cdot \vec{a}) \vec{a}}{|\vec{a}|^2} = \frac{4}{3} (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow \frac{\{(\lambda \hat{i} - 3\hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} - \hat{k})\} (\hat{i} - \hat{j} - \hat{k})}{(1 + 1 + 1)}$$

$$= \frac{4}{3} (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow (\lambda + 3 - 1) (\hat{i} - \hat{j} - \hat{k}) = 4 (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow (\lambda + 2) (\hat{i} - \hat{j} - \hat{k}) = 4 (\hat{i} - \hat{j} - \hat{k})$$

On equating the coefficient of $\hat{i}$, we get

$$\lambda + 2 = 4 \Rightarrow \lambda = 2$$

3. Ans (20)

Since, the given line touches the given circle, the length of the perpendicular from the centre $(2, 4)$ of the circle to the line $3x - 4y - k = 0$ is equal to the radius $\sqrt{4 + 16 + 5} = 5$ of the circle.

$$\therefore \frac{3 \times 2 - 4 \times 4 - k}{\sqrt{9 + 16}} = \pm 5$$

$$\Rightarrow k = 15 \quad [\because k > 0]$$

hence equation of tangent is

$$3x - 4y - 15 = 0 \quad \dots (1)$$

Let equation of normal to circle $4x + 3y = \lambda$

It passes through centre $(2, 4) \Rightarrow \lambda = 20$

hence equation of normal is

$$4x + 3y = 20 \quad \dots (2)$$

Solve (1) & (2)

$$a = 5, b = 0; k + a + b = 15 + 5 + 0 = 20$$

4. Ans (3)

$$\text{Shortest distance} = \frac{|\vec{a} - \vec{c} \cdot \vec{b} \times \vec{d}|}{|\vec{b} \times \vec{d}|}$$

$$\text{Here } \vec{a} = 3\hat{i} + 8\hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} - \hat{j} + \hat{k}, \vec{c} = -3\hat{i} - 7\hat{j} + 6\hat{k}, \vec{d} = -3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\therefore \text{Shortest distance} = \frac{270}{\sqrt{270}} = \sqrt{270} = 3\sqrt{30} = \lambda\sqrt{30} \Rightarrow \lambda = 3$$

5. Ans (6)

Let the equation of the circle be

$$(x - a)^2 + (y - a)^2 = a^2, a > 0,$$

It touches $4x + 3y - 12 = 0$

$$\therefore \left| \frac{4a + 3a - 12}{5} \right| = a$$

$$7a - 12 = \pm 5a$$

take '+' sign take '-' sign

$$2a = 12 \quad 12a = 12$$

$$a = 6 \quad a = 1$$

$\therefore$ radius of larger circle $a = 6$